

BrightLearn Research Assignment: The Language of Business Analytics:

Foundational Concepts of Analytics:

1) Business Analytics:

Business Analytics is the use of data to understand how a business is performing and to help make better decisions.

Real World Example:

A store would look at its past sales data to decide which products to stock more of during busy periods.

Source: Davenport, T. H., & Harris, J. G. (2001). *Competing on Analytics*. Harvard Business School Press.

2) Data Analytics:

Data Analytics is the process of examining data to find useful information and patterns. This helps businesses understand what is happening & make better decisions based on facts.

Real World Example:

A company will analyze customer data to see which products are selling the most and uses that information to improve its marketing strategy.

Source:

Provost, F., & Fawcett, T. (2013). Data Science for Business. O'Reilly Media.

3) Business Intelligence (BI)

Business Intelligence is the process of collecting & analyzing data to help businesses make better decisions. It uses tools & dashboards to turn raw data into clear and useful information.

Real World Example:

A company uses a BI tool like Power BI to create a dashboard that shows sales performance across different areas.

Source: Negash, S. (2004). Business Intelligence. Communications of the Association for Information Systems.

4) Data-Driven Decision Making:

Data-driven decision making means using data and facts to guide business decisions instead of relying only on guesswork. It helps businesses make more accurate & informed choices.

Real World Example:

A company uses customer data to decide which products to promote, instead of just choosing based on opinion.

Source: Provost, F., & Fawcett, T. (2013). Data Science for Business. O'Reilly Media.

5) Big Data:

Big Data refers to very large amounts of data that are too complex to handle using normal tools. It includes data that comes in quickly and in different formats, so special technologies are needed to store and analyze it.

Real World Example:

A company like Netflix analyzes large amounts of user data, such as what people watch and how long they watch, to recommend shows.

Source: Mayer-Schönberger, V., & Cukier, K. (2013). Big Data. Houghton Mifflin Harcourt.

6) Descriptive Analytics:

Descriptive analytics looks at past data to show what has already happened in a business. It helps summarize information using reports, charts and dashboards.

Real World Example:

A company creates a monthly sales report to see how much was sold in the previous month.

Source: Gartner (2023). Analytics Ascendancy Model.

7) Diagnostic Analytics:

Diagnostic analytics looks at data to understand why something happened. It helps find the reasons behind changes in performance by comparing and exploring different data points.

Real World Example:

A company notices a drop in sales and analyzes the data to find that it was caused by higher product prices.

Source: Gartner (2023). Analytics Ascendancy Model.

8) Predictive Analytics:

Predictive analytics uses past data to make estimates about what might happen in future. It looks for patterns and trends to help businesses plan ahead.

Real World Example:

A retail store predicts future demand for products based on previous sales data.

Source: Shmueli, G., et al. (2017). Data Mining for Business Analytics. Wiley.

9) Prescriptive Analytics:

Prescriptive analytics uses data to suggest the best actions a business should take. It helps decision-makers choose the most effective option based on different possible outcomes.

Real World Example:

A delivery company uses data to decide the fastest and cheapest routes for drivers.

Source: Davenport, T. H. (2015). The AI Advantage. MIT Press.

10) Data Literacy:

Data Literacy is the ability to read, understand and use data correctly. It means being able to interpret data and make sense of what is showing.

Real World Example:

An employee reads a sales chart and understands which product is performing best without needing help.

Source: Mandinach, E. B., & Gummer, E. S. (2016). Data Literacy for Educators. Teachers College Record.

② Data, Measurement & Reporting:

11) Structured Data:

Structured data is data that is organised in a clear format, usually in rows and columns like a table. It is easy to store, search and analyse as well.

Real World Example:

A company stores customer information like names, phone numbers and purchase history in a database table.

Source: Provost, F., & Fawcett, T. (2013). Data Science for Business. O'Reilly Media.

12) Unstructured Data:

Unstructured data is information that does not have a fixed format or organisation. It is harder to analyse because it includes things like text, images and videos.

Real World Examples:

Customer reviews written on social media or product review websites.

Source: Mayer-Schönberger, V., & Cukier, K. (2013). Big Data. Houghton Mifflin Harcourt.

13) Data Warehouse:

A data warehouse is a central place where a company stores large amounts of data from different sources. It is organised so that the data can be easily analysed for reporting and decision-making.

Real World Example:

A retail company combines sales data from all its stores into one system to analyse overall performance.

Source: Inmon, W. H. (2005). Building the Data Warehouse. Wiley.

14) Data Mining:

Data Mining is the process of searching through large amounts of data to find patterns, trends, or useful information. It helps businesses discover insights they might not notice easily.

Real-World Example:

A supermarket finds that customers who buy bread often also buy milk, so they place them closer together.

Source: Han, J., Kamber, M., & Pei, J. (2011). Data Mining: Concepts and Techniques. Morgan Kaufmann.

15) Data Visualization:

Data visualization is the process of showing data in a visual way like charts, graphs or maps. It helps people understand information quickly and easily.

Real World Example:

A company uses a bar chart to compare monthly sales across different branches.

Source: Few, S. (2009). Now You See It. Analytics Press.

16) Dashboard:

A dashboard is a visual tool that shows important business data in one place. It helps people quickly track performance using charts, graphs and key numbers.

Source: Eckerson, W. (2010). Performance Dashboards. Wiley.

17) Metric:

A metric is a specific number or measurement used to track performance in a business. It helps show how well something is doing over time.

Real World Example:

A website tracks the number of visitors each day as a metric to measure traffic.

Source: Parmenter, D. (2015). Key Performance Indicators, Wiley.

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18) Key Performance Indicator (KPI):

A KPI is a specific measure used to check how well a business or team is achieving its goals. It helps track progress and shows whether performance is improving or not.

Real World Example:

A marketing team uses the number of new customers gained each month as a KPI.

Source: Parmenter, D. (2015). Key Performance Indicators. Wiley.

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19) Benchmark:

A benchmark is a standard or reference point used to compare performance. Businesses use it to see how well they are doing compared to others or against a target.

Real-World Example:

A company compares its customer satisfaction score to the industry average to see if it is performing well.

Source: Investopedia (2023). Benchmark Definition and Meaning.

20) Segmentation:

Segmentation is the process of dividing customers or data into smaller groups based on shared characteristics. This helps businesses understand different types of customers and target them more effectively.

Real World Example:

A clothing brand groups customers by age and buying habits to send more personalised promotions.

Source: Kotler, P., & Keller, K.L. (2016). Marketing Management. Pearson.

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③ Customer & Sales KPIs:

21) Conversion Rate:

Conversion rate is the percentage of people who complete a desired action after interacting with a business, such as buying a product or signing up. It shows how effective a website, advert or campaign is at getting results.

Real World Example:

If 100 people visit an online store and 5 of them make a purchase, the conversion rate is 5%.

Source: Chaffey, D. (2019). Digital Marketing. Pearson.

22) Customer Acquisition Cost (CAC) :
Customer Acquisition Cost is the amount of money a business spends to get a new customer. It includes costs like advertising, marketing and sales efforts.

Real World Example:

A company spends ₹10,000 on ads and gains 100 new customers, so the CAC is ₹100 per customer.

Source: Investopedia (2023). Customer Acquisition Cost (CAC) Definition.

23) Customer Lifetime Value (CLV / LTV) :

Customer Lifetime Value is the total amount of money a business expects to earn from a customer over the whole time they stay with the company. It helps businesses understand how valuable each customer is in the long run.

Real World Example:

Gupta, S., & Lehmann, D. (2003). Customers as Assets. Journal of Interactive Marketing.

24) Churn Rate:

Churn rate is the percentage of customers who stop using a product or service over a certain period of time. It helps businesses understand how many customers they are losing.

Real World Example:

A streaming service tracks how many users cancel their subscriptions each month to measure customer loss.

Source: Reicheld, F.F. (1996). *The Loyalty Effect*.
Harvard Business School Press.

25) Customer Retention Rate:

This refers to the percentage of customers a business keeps over a certain period of time. It shows how well a company is able to keep its customers coming back.

Real World Example:

A gym tracks how many members renew their memberships each year to see how many customers they retain.

Source: Reichheld, F.F. (1996). *The Loyalty Effect*.
Harvard Business School Press.

26) Net Promoter Score (NPS) :
Net Promoter Score is a measure of how likely customers are to recommend a business to others. It is based on a simple question and helps show customer satisfaction and loyalty.

Real World Example :

A company asks customers to rate, from 0 to 10, how likely they are to recommend their service, then uses the results to measure customer loyalty.

Source : Reichheld, F.F. (2003). The One Number You Need to Grow. Harvard Business Review.

27) Average Order Value (AOV) :

Refers to the average amount of money a customer spends each time they place an order. It helps businesses understand how much customers typically spend per purchase.

Real World Example :

If a store makes ₹5000 from 100 orders, the average order value is ₹50.

Source : Chaffey, D. (2019). Digital Marketing. Pearson.

28) Sales Funnel:

A sales funnel shows the journey a customer takes from first hearing about a product to finally making a purchase.

Real World Example:

A customer sees an ad, visits a website, compares products and then buys.

Source: Kotler, P., & Keller, K. L. (2016). Marketing Management. Pearson.

④ Marketing & Campaign Analytics:

29) Marketing Campaign:

A marketing campaign is a planned set of activities used to promote a product or service to a target audience. It usually runs for a specific period and uses channels like social media, email or ads.

Real World Example:

A company runs a month-long social media campaign to promote ~~an~~ a new product, using ads, influencer posts and special discounts.

Source: Kotler, P., & Keller, K. L. (2016). Marketing Management. Pearson.

30) Post-Campaign Analysis :

This refers to the process of reviewing and evaluating a marketing campaign after it has finished. It helps businesses understand what worked well and what can be improved in the future.

Real World Example :

After a social media campaign ends, a company checks metrics like clicks, sales and engagement to see if the campaign was successful.

Source : Chaffey, D. (2019). Digital Marketing. Pearson.

31) Incremental Revenue :

Incremental Revenue refers to the extra income a business earns from a specific action or campaign.

Real World Example :

A company runs a promotion and sees sales increase by ₹20,000 compared to normal sales. This extra amount is the incremental value.

Source : Kotler, P., & Keller, K.L. (2016). Marketing Management. Pearson.

32) Return on Investment (ROI):

Return on Investment is a measure of how much profit a business makes from an investment compared to how much it cost.

Real World Example:

A company spends ₹5000 on a marketing campaign and earns ₹10,000 from it, showing a positive return.

Source: Investopedia (2023). Return on Investment (ROI) Definition.

33) Return on Ad Spend (ROAS):

This measures how much revenue a business earns for every amount spent on advertising. It shows how effective an ad campaign is at generating sales.

Real World Example:

If a company spends ₹1000 on ads and earns ₹4000 in sales, the ROAS is 4:1.

Source: Chaffey, D. (2019). Digital Marketing. Pearson.

34) Click-Through Rate (CTR) :
CTR refers to the percentage of people who click on a link or advert after seeing it. It shows how effective an ad is at getting attention.

Real World Example :

If 1000 people see an online advert and 50 click on it, the CTR is 5%.

Source : Chaffey, D. (2019). Digital Marketing. Pearson.

35) Bounce Rate :

Bounce rate refers to the percentage of people who visit a website and leave without clicking on anything.

Real World Example :

If 100 people visit a website and 60 leave after viewing only one page, the bounce rate is 60%.

Source : Chaffey, D. (2019). Digital Marketing. Pearson.

36) A/B Testing:

A/B testing is when two versions of something are compared to see which one performs better.

Real World Example:

A company tests two different email subject lines to see which one gets more people to open the email.

Source: Kohavi, R., & Longbotham, R. (2011).

Online Controlled Experiments and A/B Testing. Cambridge University Press.

37) Lead Generation:

Lead generation is the process of attracting and collecting contact details from potential customers who are interested in a product or service.

Real World Example:

A company offers a free ebook on its website in exchange for a visitor's email address.

Source: Kotler, P., & Keller, K.L. (2016). Marketing Management. Pearson.

38) Marketing Attribution:
Refers to the process of identifying which marketing channels contributed to a sale.

Real World Example:
A customer sees a Facebook ad and then clicks a Google ad later and makes a purchase, attribution helps decide which channel gets credit for the sale.

Source: Chaffey, D. (2019). Digital Marketing. Pearson.